INCH-POUND
MIL-C-55302/118C
31 October 1988
SUPERSEDING
MIL-C-55302/118B(USAF)
12 April 1983

MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, DECADE INCREMENT 20 THRU 110 AND 128 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.050 SPACING)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.

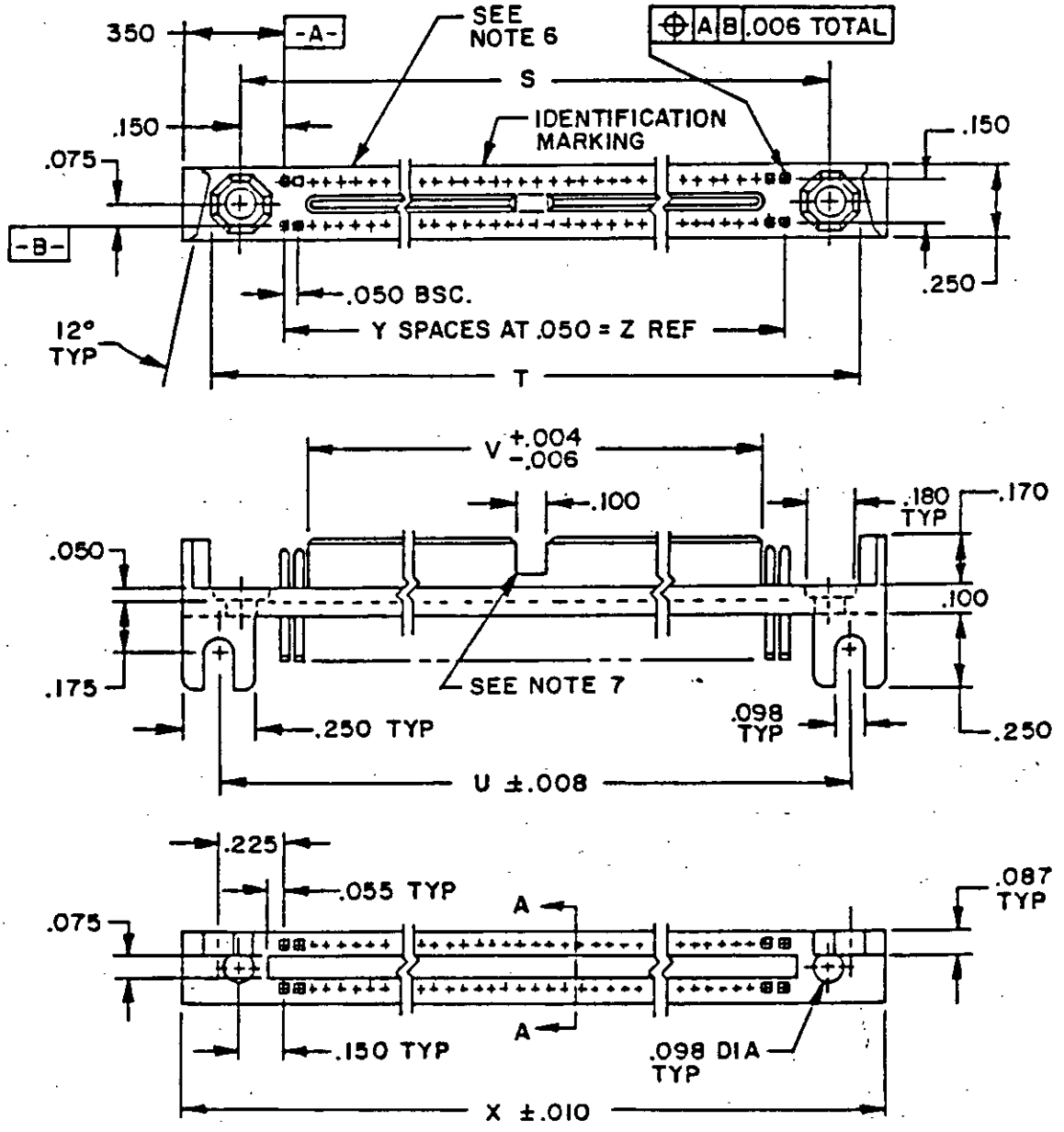
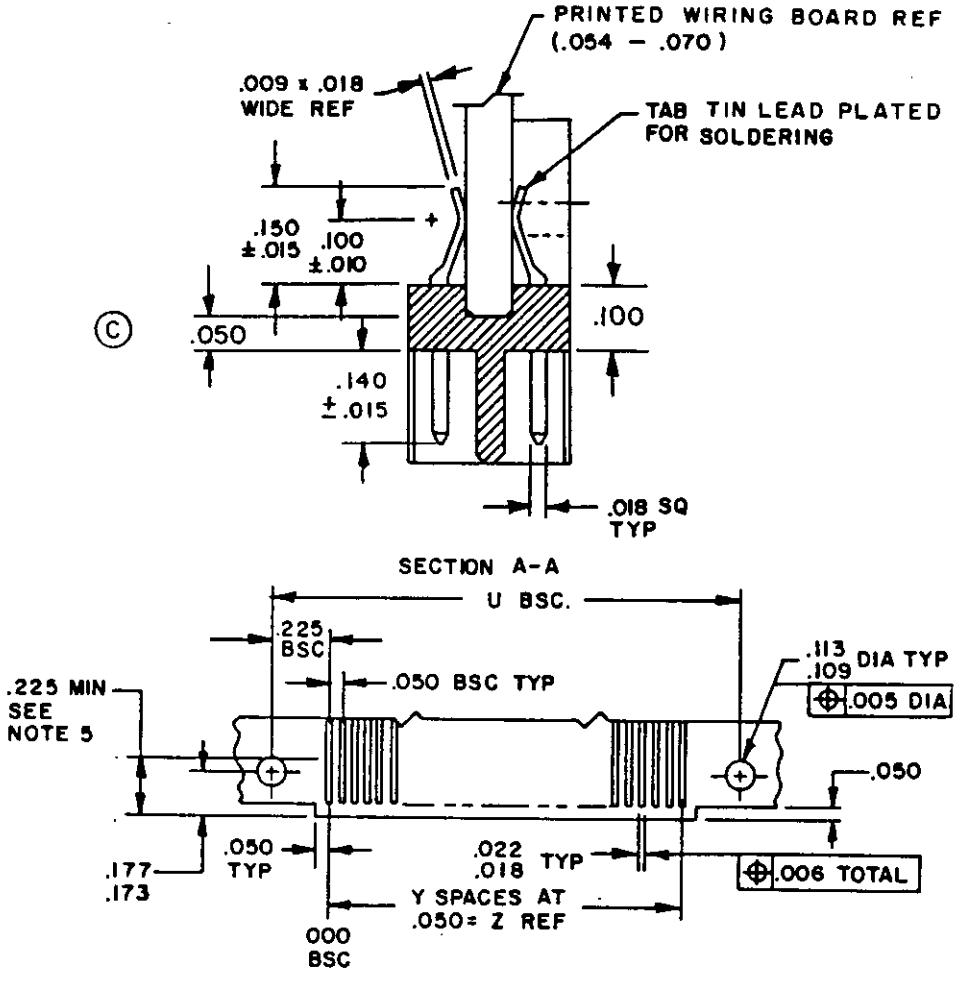


FIGURE 1. Connectors, plug without pin protectors .050 (1.27 mm) spacing for style 1.

© denotes changes

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RECOMMENDED PRINTED WIRING BOARD LAYOUT

Inches	mm	Inches	mm	Inches	mm
.004	.10	.050	1.27	.113	2.87
.005	.13	.054	1.37	.140	3.56
.006	.15	.055	1.40	.150	3.81
.008	.20	.070	1.78	.170	4.32
.009	.23	.075	1.90	.173	4.39
.010	.25	.087	2.21	.175	4.44
.015	.38	.098	2.49	.177	4.50
.018	.46	.100	2.54	.180	4.57
.022	.56	.109	2.77	.225	5.72
				.250	6.35
				.350	8.89

FIGURE 1. Connectors, plug without pin protectors .050 (1.27 mm) spacing for style 1 - Continued.

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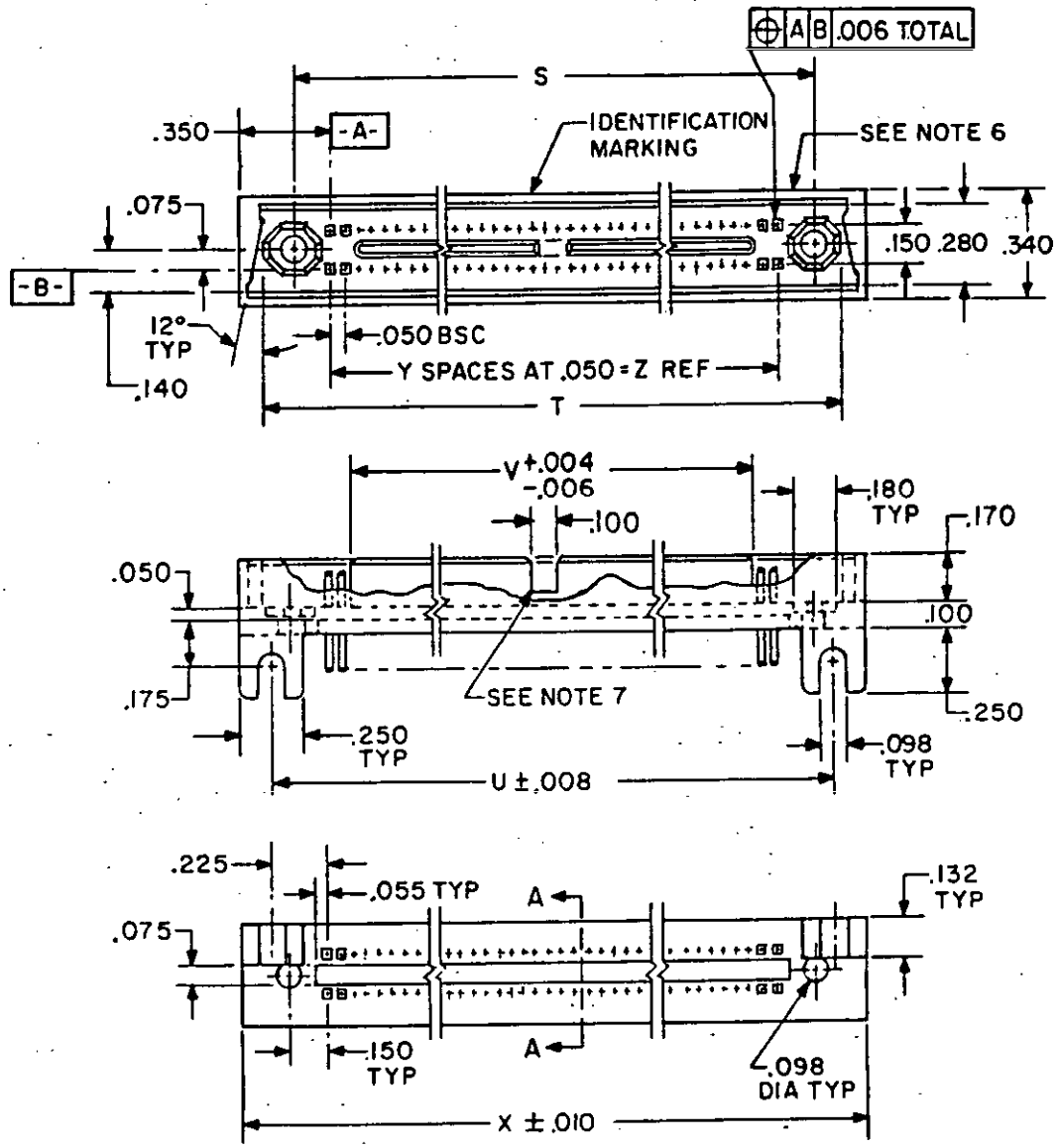
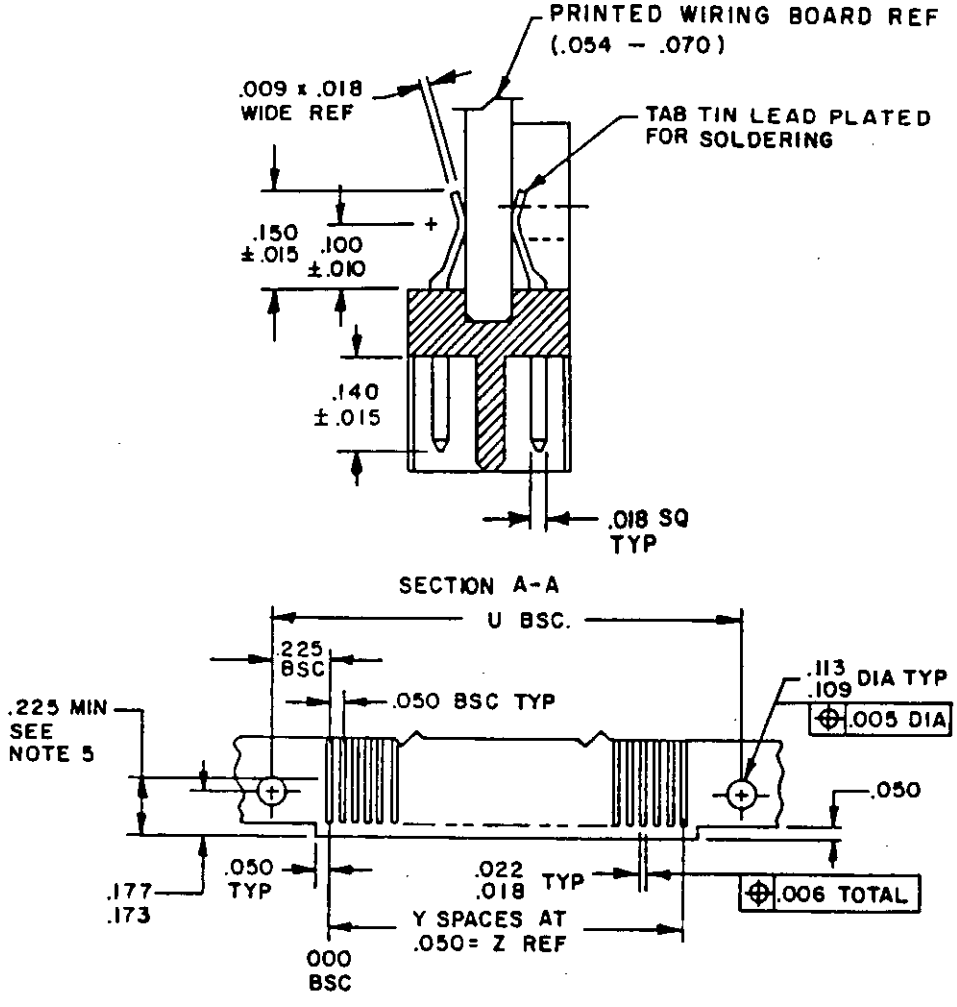


FIGURE 2. Connectors, plug with pin protectors .050 (1.27 mm) spacing for style 2.

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RECOMMENDED PRINTED WIRING BOARD LAYOUT

Inches	mm	Inches	mm	Inches	mm
.004	.10	.054	1.37	.150	3.81
.005	.13	.055	1.40	.170	4.32
.006	.15	.070	1.78	.173	4.39
.008	.20	.075	1.90	.175	4.44
.009	.23	.098	2.49	.177	4.50
.010	.25	.100	2.54	.180	4.57
.015	.38	.109	2.77	.225	5.71
.018	.46	.113	2.87	.250	6.35
.022	.56	.132	3.35	.280	7.11
.050	1.27	.140	3.56	.340	8.64
				.350	8.89

FIGURE 2. Connectors, plug with pin protectors .050 (1.27 mm)spacing for style 2 - Continued.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm) and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/119(USAF).
5. Minimum pad length on printed wiring board.
6. Numbers indicating end positions stamped this side only.
7. Except 20 through 50 position.

Table I. Dash numbers and dimension. (Style 1)

Dash number	Number of contacts	Dimensions							Contact identification numbers
		S	T	U	V	X	Y	Z	
01	20	.750 (19.05)	.970 (24.64)	.900 (22.86)	.290 (7.37)	1.150 (29.21)	9	.450 (11.43)	1-10
02	30	1.000 (25.40)	1.220 (30.99)	1.150 (29.21)	.540 (13.72)	1.400 (35.56)	14	.700 (17.78)	1-15
03	40	1.250 (31.75)	1.470 (37.34)	1.400 (35.56)	.790 (20.07)	1.650 (41.91)	19	.950 (24.13)	1-20
04	50	1.500 (38.10)	1.720 (43.69)	1.650 (41.91)	1.040 (26.42)	1.900 (48.26)	24	1.200 (30.48)	1-25
05	60	1.750 (44.45)	1.970 (50.04)	1.900 (48.26)	1.290 (32.77)	2.150 (54.61)	29	1.450 (36.83)	1-30
06	70	2.000 (50.80)	2.220 (56.39)	2.150 (54.61)	1.540 (39.12)	2.400 (60.96)	34	1.700 (43.18)	1-35
07	80	2.250 (57.15)	2.470 (62.74)	2.400 (60.96)	1.790 (45.47)	2.650 (67.31)	39	1.950 (49.53)	1-40
08	90	2.500 (63.50)	2.720 (69.09)	2.650 (67.31)	2.040 (51.82)	2.900 (73.66)	44	2.200 (55.88)	1-45
09	100	2.750 (69.85)	2.970 (75.44)	2.900 (73.66)	2.290 (58.17)	3.150 (80.01)	49	2.450 (62.23)	1-50
10	110	3.000 (76.20)	3.220 (81.79)	3.150 (80.01)	2.540 (64.52)	3.400 (86.36)	54	2.700 (68.58)	1-55
11	128	3.450 (87.63)	3.670 (93.22)	3.600 (91.44)	2.990 (75.95)	3.850 (97.79)	63	3.150 (80.01)	1-64

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Table II. Dash numbers and dimensions. (Style 2)

Dash number	Number of contacts	Dimensions							Contact identification numbers
		S	T	U	V	X	Y	Z	
12	20	.750 (19.05)	.970 (24.64)	.900 (22.86)	.290 (7.37)	1.150 (29.21)	9	.450 (11.43)	1-10
13	30	1.000 (25.40)	1.220 (30.99)	1.150 (29.21)	.540 (13.72)	1.400 (35.56)	14	.700 (17.78)	1-15
14	40	1.250 (31.75)	1.470 (37.34)	1.400 (35.56)	.790 (20.07)	1.650 (41.91)	19	.950 (24.13)	1-20
15	50	1.500 (38.10)	1.720 (43.69)	1.650 (41.91)	1.040 (26.42)	1.900 (48.26)	24	1.200 (30.48)	1-25
16	60	1.750 (44.45)	1.970 (50.04)	1.900 (48.26)	1.290 (32.77)	2.150 (54.61)	29	1.450 (36.83)	1-30
17	70	2.000 (50.80)	2.220 (56.39)	2.150 (54.61)	1.540 (39.12)	2.400 (60.96)	34	1.700 (43.18)	1-35
18	80	2.250 (57.15)	2.470 (62.74)	2.400 (60.96)	1.790 (45.47)	2.650 (67.31)	39	1.950 (49.53)	1-40
19	90	2.500 (63.50)	2.720 (69.09)	2.650 (67.31)	2.040 (51.82)	2.900 (73.66)	44	2.200 (55.88)	1-45
20	100	2.750 (69.85)	2.970 (75.44)	2.900 (73.66)	2.290 (58.17)	3.150 (80.01)	49	2.450 (62.23)	1-50
21	110	3.000 (76.20)	3.220 (81.79)	3.150 (80.01)	2.540 (64.52)	3.400 (86.36)	54	2.700 (68.58)	1-55
22	128	3.450 (87.63)	3.670 (93.22)	3.600 (91.44)	2.990 (75.95)	3.850 (97.79)	63	3.150 (80.01)	1-64

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figures 1 and 2.

Material:

Body: In accordance with MIL-C-55302.

© Plating: Contact plating shall be gold over nickel in accordance with MIL-C-55302 in the contact engagement area for .100 inch minimum length. The termination area plating shall be tin lead over nickel per MIL-C-55302 for .145 inch minimum length. The remainder of the contact shall be plated with nickel per MIL-C-55302.

Contact identification: See figures 1 and 2.

Contact rating: 1.5 amperes per contact, maximum.

Operating temperature: -65°C to 125°C.

Keying: See MIL-C-55302/31.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .30 and the withdrawal force in pounds shall be a minimum of .03 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: The average contact resistance of all contacts measured shall not exceed .015 ohm, and no individual contact pair shall have a resistance exceeding .020 ohm.

Contact retention: .75 pound, minimum.

Dielectric withstanding voltage:

Sea level: 600 volts rms.

High altitude: 150 volts rms.

Solderability: Each connector shall be subjected to method 208 of MIL-STD-202 (solder termination only).

Part number: M55302/118- (dash number from tables I and II).

Patent number 3,404,367. The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

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Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Air Force - 11, 85, 99
DLA - ES

User activities:
Navy - SH, AS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - CR

Agent:
DLA - ES

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